SIMULATING LOWER LIMB PROSTHETIC USE FOR DATA-DRIVEN CARE: PROSTHETIC OUTCOMES ANALYSIS

Drexel University

DSCI 611 — Winter 2026

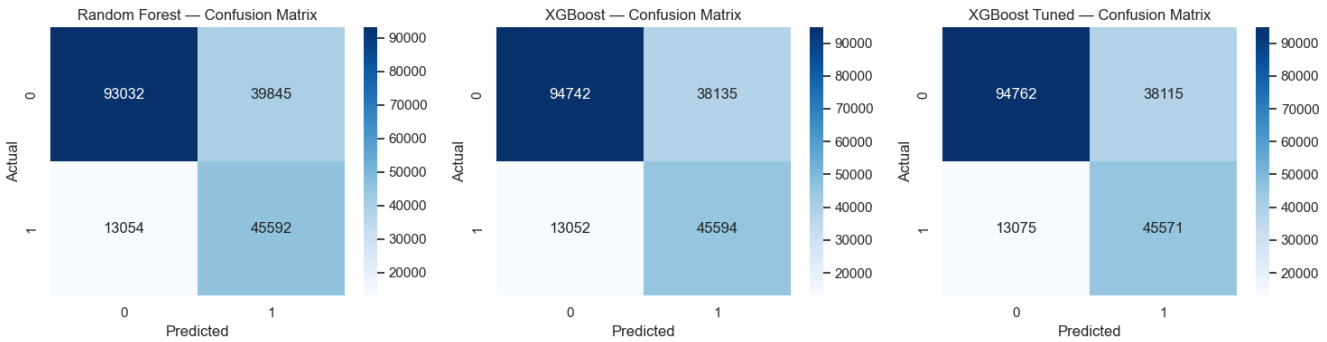
Michael Huang
Virginia Muthard
Valentina Ozornina

March 15, 2026

Executive Summary

The purpose of this project is to explore the use of workflow automation in prosthetic component selection. The project uses a synthetic, randomly-generated dataset of one million prosthetic users containing features that cover various details included in the prosthetic process–such as patient date of birth, date of amputation, side and level of amputation, clinician information, and functional outcome score with the device. The primary research question this project aims to answer is: Can an automated workflow using reproducible and replicable modeling be created to improve efficiency in determining prosthetic success? The focus of the project revolves around generating reproducible pipelines for the analysis of data rather than identifying predictors of prosthetic success. No interpretable conclusions about predictive features of prosthetic success will be made from this project; introduction of real-world data in the future could serve as a basis for model evaluation in an independent study and is outside the scope of this project.

Some feature engineering was performed on the dataset, including the engineering of a binary classification target of 'positive/negative' for `outcome_score`. Given the randomness in the original data, no signaling for prediction was present to provide feedback on the automation of the model. However, further engineering of the target created a synthetic target which reflected the $\sim 31\%$ positive class imbalance. This generated a weak signal and moved the model performance away from near-chance predictions. The models chosen for analysis included classifier models based on Random Forest and XGBoost. Initial evaluation on the models was performed using primarily ROC-AUC and PR-AUC metrics. These metrics were specifically chosen due to the imbalanced binary classification problem present within the data. More specifically, these metrics do not just explore prediction success; rather, they consider how frequently the model is performing successfully across the predictions. With the XGBClassifier model boasting slightly better performance in these metrics, it was chosen for hyperparameter tuning and final model evaluation.



Confuusion matrices for RFClassifier, XGBClassifier, and tuned-XGBClassifier models

Hyperparameter tuning was completed using a custom script to explore random combinations for the parameters `n_estimators`, `learning_rate`, `max_depth`, `subsample`, `colsample_bytree`, and `reg_lambda`. A validation set was created from the train data, and the models were retrained using the best parameters derived from the tuning. Model performance was re-evaluated with an increase in performance of the tuned XGBClassifier model of $< 0.05\%$. This result demonstrated a minimal performance improvement over the untuned RFClassifier and XGBClassifier models. In the experiments, the greatest performance yield came from the generation of the weak-signal synthetic target over baseline modeling using the original target. The project results demonstrated that an automated workflow can be used to determine weak predictive signaling in prosthetic data. In conclusion, the automated workflow built within this project serves as proof-of-concept for increasing efficiency in prosthetic component selection, deeming the project successful. With the introduction of real-world data, the workflow could be evaluated in a new study about predictors of prosthetic success that could yield usable clinical interpretations.

SIMULATING LOWER LIMB PROSTHETIC USE FOR DATA-DRIVEN CARE: PROSTHETIC OUTCOMES ANALYSIS

M. Huang[1], **V. Muthard**[1] **and V. Ozornina**[1]

[1] *Drexel University*

DSCI 611 — Winter 2026

Abstract

This study generates a reproducible workflow for the analysis of prosthetic outcome success. The datasets used in this study have been synthetically-generated for use in domain-based educational exploration. Predictive models are proposed and examined for integration into an automated pipeline for tuning and analysis of metrics. XGBoost Regression and Classification models are used in the analysis, followed by hyperparameter tuning. Results demonstrate that the workflow is reproducible and able to accept real-world data for assessing the predictive power of the dataset features, which is beyond the scope of this project.

Contents

Executive Summary i
1 Introduction 1
1.1 Background 1
1.2 Goals and Objectives 1
1.3 Stakeholders 1
2 Data Sources and Data Preparation 2
2.1 Data Background and Sourcing 2
2.2 Prosthetic User Dataset 2
2.3 Prosthetic Component Datasets 2
2.4 Healthcare Provider Datasets 2
2.5 Dataset Scope & Limitations 2
3 Data Exploration and Cleaning 2
3.1 Exploratory Analysis 2
3.2 Data Cleaning and Feature Engineering 3
4 Methodology and Modeling 4
4.1 Definition of Successful Outcome 4
4.2 Methodology 4
4.3 Model Selection 4
5 Results 4
5.1 Evaluation Metrics 4
5.2 RandomForestClassifier Performance 4
5.3 XGBClassifier Performance 4
5.4 Hyperparameter Tuning 5
5.5 Final Model Evaluation 5
6 Discussion 5
7 Conclusion and Future Work 5
8 Reflection 5
A Repository 7
B Pipeline Stages 7
C AI Disclosure 7

Scope and Definitions

This study analyzes synthetically-derived, lower limb prosthetic data under a defined analytical scope using reproducible automation. Temporally, the data is restricted to current prosthetic usage; no historical records exist for users. All data and analyses are strictly for educational purposes and do not directly reflect any real-world data. The use of any manufacturer or provider name along with any associated findings does not represent, recommend, or endorse any specific manufacturer or component for use.

1. Introduction

1.1. Background

Patients undergoing lower limb amputation suffer significantly diminished ambulatory function among other adverse functional limitations [1]. These individuals are reliant on healthcare providers to make appropriate clinical decisions regarding prosthetic componentry for good functional outcomes [5]. Currently, there is no robust evidence base to guide component selection for prosthetic users [2], [3]. As a result, the process is often time-consuming and may be influenced by clinician familiarity and personal preference rather than objective comparative evidence [4]. Some researchers have proposed leveraging data science for the process [3], which would create a more evidence-based approach to component selection. With the use of automated workflows, a machine learning task could be effectively and efficiently integrated into a clinical framework. The result would be a systematic, empirically supported approach to component selection that significantly reduces clinician burden.

1.2. Goals and Objectives

This project is serving as a subsequent step for *Simulating Lower Limb Prosthetic Use for Data-Driven Care*, a prior Drexel University data science project rooted in cloud computing and yielding a knowledge-based recommender model to improve the efficiency of prosthetic component selection. The professional objective in pursuing the current project is to generate a proof-of-concept, automated pipeline to predict prosthetic success given various personal and clinical features. The goal is not to find predictors of prosthetic success; rather, the objective of this project is to create an efficient workflow for the task, improve efficiency with large data analysis, and mitigate errors in data processing while creating a reproducible and replicable model workflow that can be used with real-world data. Specifically, the goals of this project are:

1. Initialization of a repository tracked with DVC.
2. Integration of data with a reproducible workflow.
3. Exploration of data using automation.
4. Predictive analysis with model training using staged inputs and outputs.
5. Hyperparameter tuning of model pipeline using metrics.

1.3. Stakeholders

Users of this automated workflow at the current stage would include academics or ML researchers with access to domain-level data to assess the accuracy and predictive power of the model itself. Future

target users include prosthetic clinics and associated clinicians to aid in more evidence-based component decisions for patients.

2. Data Sources and Data Preparation

2.1. Data Background and Sourcing

The analysis incorporates six prosthetic datasets aggregated into a single, combined dataset for completion of the project goals. All datasets were synthetically and randomly created to include features associated with prosthetic care and patient management due to the non-existence of public data containing this information from both HIPAA and intellectual property perspectives. Previously, the datasets were used as a foundation for a knowledge-based recommender system project, as well as the basis for a database construction project while enrolled at Drexel University.

2.2. Prosthetic User Dataset

sim_prosthetics_data_expanded.csv This dataset is the primary dataset for analysis into which the secondary datasets are merged. The data includes 1M simulated prosthetic patient records cataloging various details about the patient and their prosthesis, including demographic, surgical, and component information. The data columns and descriptions are documented in Table 1.

Table 1. Data Columns in `sim_prosthetics_data_expanded.csv`.

Column	Type	Description
`patient_id`	str	Unique identifier for each prosthetic user.
`dob`	obj	Date of birth of user.
`gender`	str	User gender.
`amputation_level`	str	Amputation level for user.
`amputation_side`	str	Amputation side for user.
`foot_mfg`	str	Manufacturer of current prosthetic foot.
`foot_mdl`	str	Model name of current prosthetic foot.
`knee_mfg`	str	Manufacturer of current prosthetic knee.
`knee_mdl`	str	Model name of current prosthetic knee.
`hip_mfg`	str	Manufacturer of current prosthetic hip.
`hip_mdl`	str	Model name of current prosthetic hip.
`fitting_date`	obj	Date user was fit with current prosthesis.
`outcome_score`	int	Functional outcome score with prosthesis (0-100.0)
`satisfaction_rating`	int	User satisfaction rating with prosthesis (0-5.0)
`clinician_id`	str	Unique identifier for each prosthetic clinician.
`surgeon_id`	str	Unique identifier for each surgeon.
`prx_material`	str	Material construction of prosthesis.
`visits_list`	list	List of follow-up visits for prosthesis.
`amputation_date`	obj	Date of amputation.

2.3. Prosthetic Component Datasets

The three prosthetic component datasets provide manufacturer and model information for the three major component levels (feet, knees, and hips). The datasets contain a random sub-sampling of components from three major prosthetic component manufacturers–`ossur`, `ottobock`, and `proteor`. They serve as secondary datasets to the primary prosthetic user dataset. Within each dataset, each component model is a unique entity, identified by an ID number to support the inclusion of more model-specific information for an individual component in the future. The three datasets include:

- ***sim_prosthetics_feet_data.csv*** Information for prosthetic feet
- ***sim_prosthetics_knee_data.csv*** Information for prosthetic knees
- ***sim_prosthetics_hip_data.csv*** Information for prosthetic hips

2.4. Healthcare Provider Datasets

The two healthcare provider datasets provide synthetic contact and provider information for the prosthetic clinicians and surgeons referenced in the data. They serve as secondary datasets to the primary prosthetic user dataset. Within each dataset, each provider is a unique entity, identified by an ID number along with contact information. The two datasets include:

- ***sim_directory_clinicians.csv*** Information for prosthetic clinicians
- ***sim_directory_surgeons.csv*** Information for surgeons

2.5. Dataset Scope & Limitations

The data represent only three levels of lower limb amputation and only three major manufacturers of prosthetic components. A random subset of component models from each manufacturer was utilized and therefore does not represent the entire portfolio for any given manufacturer. Additionally, synthetic clinician and surgeon entities represent only a small geographical subset of care. Together with the random generation of the data and lack of true 'ground truth', discovering key indicators for predictive success is not expected. Therefore, true model testing should be performed on real-world data outside this study to assess model accuracy and predictive power using the dataset features.

3. Data Exploration and Cleaning

3.1. Exploratory Analysis

All datasets were retrieved through a reproducible workflow leveraging cloud-based raw data storage, with access managed via DVC and Google Cloud Services (GCS) authentication. The repository structure, file locations, and descriptions for the project are detailed in Appendix A. Initial exploratory analysis was performed to observe dataset properties and features. Distributions of the features in the dataset were viewed to gain insight into the dataset structure and statistical properties. For example, `hip disarticulation` presented the least number of times within the dataset, as demonstrated in Figure 1. Similarly, the `outcome_score` target was discovered to be fairly normally distributed as seen in Figure 2.

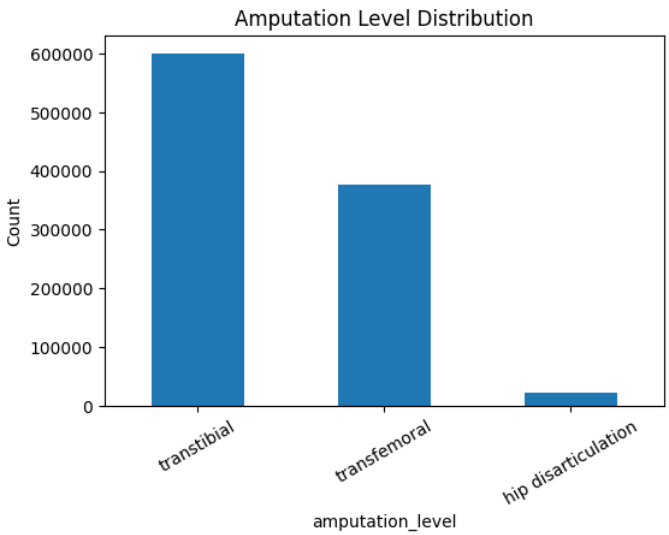


Figure 1. Distribution of amputation levels within dataset.

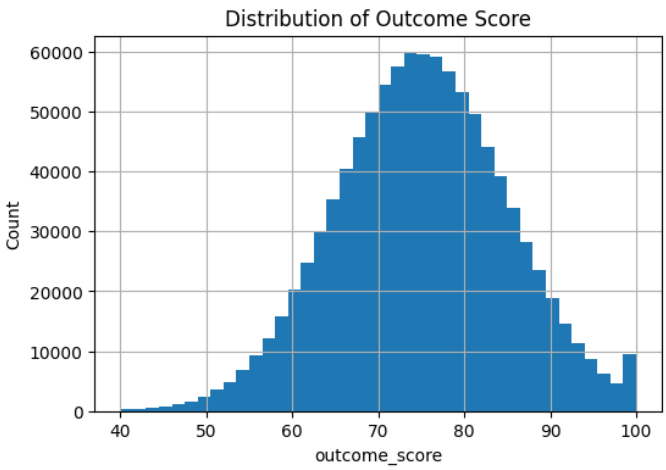


Figure 2. Distribution of outcome scores across the dataset.

The components were also examined for their contributions to the dataset. Figure 3 displays the distribution of components within the dataset broken out by manufacturer. It can be noted from this graph that while all manufacturers have counts for foot and knee components, only `ottobock` has counts for hip components. A website search for the other manufacturers confirmed that neither `ossur` nor `proteor` manufactures hip components.

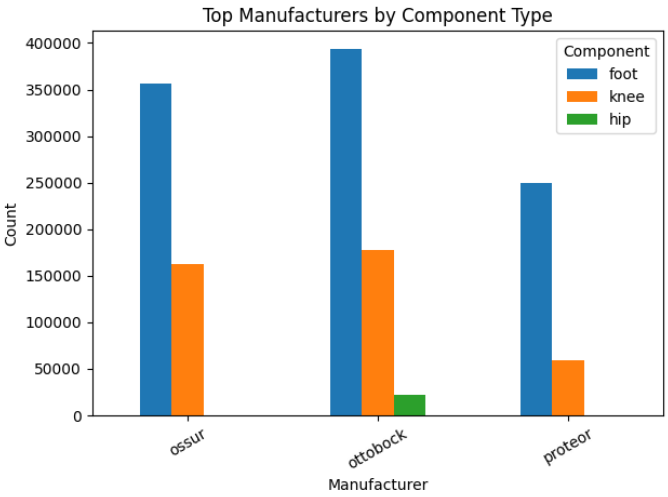


Figure 3. Distribution of components by manufacturer.

3.2. Data Cleaning and Feature Engineering

The primary dataset, `sim_prosthetics_data_expanded.csv`, presented a fairly clean starting basis for pre-processing as did the secondary datasets. Secondary datasets were merged into the primary dataset using a left-join strategy. This approach created a wide table of features that could be used in analysis. Minor cleaning was performed to ensure data types were usable for the analysis and that labeling was consistent. The presence of NaN values was also assessed to ensure appropriate handling. One confounding feature of the dataset is that while all amputation levels require a foot component, not all levels require a knee or hip component. This factor led to the majority of the NaN values present within the dataset, which can be seen in the missing values for knee and hip components in the Component Merge Quality analysis in Table 2.

Feature engineering was also performed to provide more opportunities for exploring potentially usable signals for prediction in the data. Figure 4 shows the distribution of the engineered feature `total_visits`, derived from the `visits_list` column as a numerical total count of follow-up visits.

Table 2. Component Merge Quality Analysis

Feature	Value
`rows`	1,000,000
`foot_id_match_rate`	1.0
`knee_id_match_rate`	0.399328
`hip_id_match_rate`	0.022055
`foot_id_missing`	0
`knee_id_missing`	600,672
`hip_id_missing`	977,945

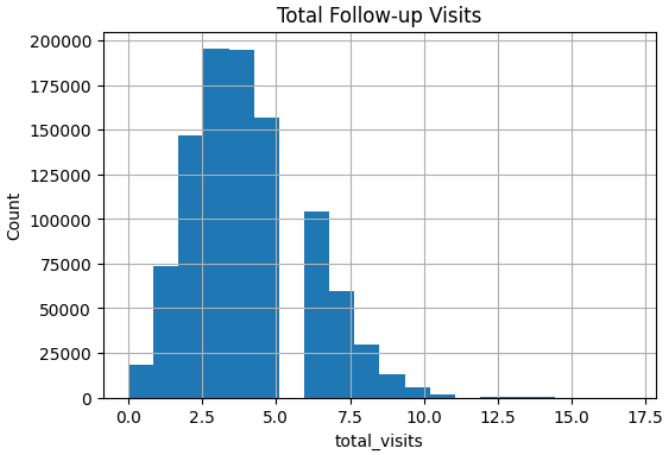


Figure 4. Distribution of total follow-up visits, engineered from the dataset.

After feature engineering, numeric variables were analyzed for interdependence. Using the correlation map in Figure 5, it was noted that true correlations only exist between derived features. This finding is expected, and confirms the low predictive signal in the dataset, re-highlighting model accuracy in prediction as a non-goal for the project. To address this limitation, the data was also explored from a classification approach. The `outcome_score` was treated as a binary target with a `outcome_score` ≥ 80 indicating a positive result.

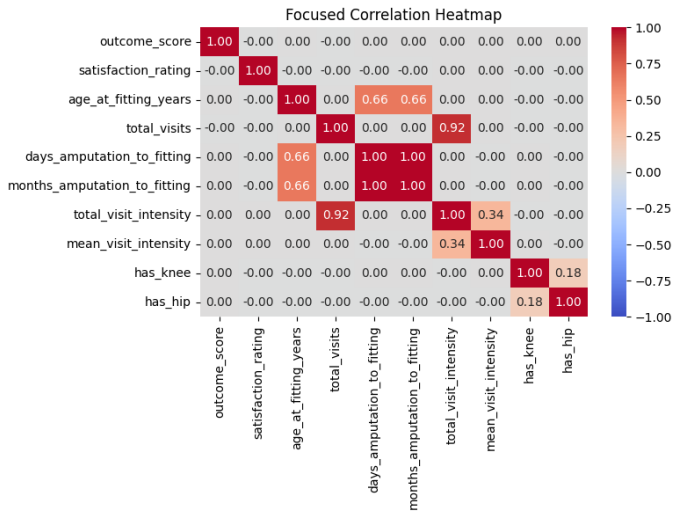


Figure 5. Correlation heatmap of numeric features.

The figures depicted in this report represent the more notable discoveries made during the data integration and exploratory data analysis. For further details of the analysis and a complete log of insights, refer to the `eda.ipynb` notebook within the repository (see Appendix A).

4. Methodology and Modeling

4.1. Definition of Successful Outcome

Given the objectives of this project, a successful outcome will include the ability to run an automated pipeline, replicating modeling results from input data. While building a predictive model remains a goal of the project, the overall project success is not defined by the predictive accuracy of the model [1]. The development of a model that is automated and reproducible–allowing analysis of real-world data–remains the primary focus.

4.2. Methodology

After full cleaning of the data, pipeline automation was used to split the data into a stratified 0.8 train/test split. With the goal of creating an automated pipeline to predict prosthetic success, `outcome_score` was selected as the original target. The initial modeling plan used a regression approach and modeling was attempted using different regression models; however, given the randomness in the data, performance was unable to improve beyond near-chance predictions. The modeling plan was then shifted to a binary classification task. The target was also changed to a generated weak-signal synthetic target, `synthetic_target`. This synthetic target is a noisy logistic function of a small number of plausible predictors (e.g., age at fitting, time-to-fitting, visit count/intensity, and amputation level indicator), calibrated to preserve a realistic class prevalence ($\sim$0.31). This decision was driven by the positive class imbalance present within the binary `outcome_score` target data, with the class distribution for the train data displayed in Table 3. This strategy ensures the task remains non-trivial while allowing meaningful evaluation of model behavior by introducing more complexity to the original target data.

Table 3. Train Target Class Imbalance

Class	Value
`positive`	31%
`negative`	69%

4.3. Model Selection

Given the expected low signal in the data, a classifier that would detect latent connections was needed. With a significant number of NaN values in the data, the classifier also needed to be robust and insensitive to their presence. For these reasons, tree-based modeling was the general goal. The decision was made to use a Random Forest (`RandomForestClassifier`) model as the first selection. Random Forest enables random subsampling within the features in the data, which can help prevent overfitting. Decisions are made by generating small forests to yield the result. This attribute was important given the low-signal in the data. The second model selected was XGBoost (`XGBClassifier`). This model relies on gradient boosting and error correction to quickly and efficiently construct trees to build predictions. In the case of XGBoost, the trees are built singularly and sequentially, with full evaluation held until the end. Both models offer similar handling with different performance tradeoffs, which is why each was selected.

5. Results

5.1. Evaluation Metrics

Given the problem presents as binary classification with a $\sim$31% positives class imbalance, selected metrics should capture both ranking quality and classification behavior to ensure results and interpretations are valid. Knowing the constraints of the data, the metrics in Table 4 were used to evaluate the chosen models. These metrics provide more insight and fairness in scoring the model's actual performance over other measures, such as accuracy, which can mask undesired behaviors.

[1] These results do not claim clinical predictive validity; they confirm pipeline correctness under controlled signal.

Table 4. Selected Evaluation Metrics

Metric	Description
`ROC-AUC`	Ranking ability across thresholds; Useful global discrimination metric; Baseline for random model $\sim$0.50
`PR-AUC`	More informative under class imbalance; Focuses on positive class performance; Baseline $\approx$ positive class prevalence ($\sim$0.31)
`Precision`	Of predicted positives, how many are correct
`Recall`	Of actual positives, how many are detected
`F1 Score`	Harmonic mean of precision and recall

5.2. RandomForestClassifier Performance

The RandomForestClassifier was trained using the `train_data` and then evaluated using the `test_data`. Predictions were observed using a confusion matrix as shown in Figure 6. The resulting `ROC-AUC` and `PR-AUC` metrics used for analysis are then displayed in Table 5.

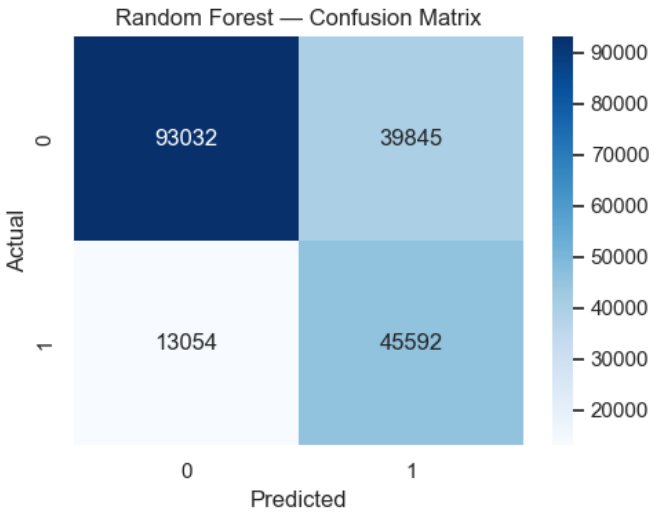


Figure 6. Confusion matrix for predictions from RandomForestClassifier.

Table 5. RandomForestClassifier Metrics Results

Metric	Synthetic Value	Original Value
`ROC-AUC`	0.503100	0.311800
`PR-AUC`	0.815636	0.647572

5.3. XGBClassifier Performance

The XGBClassifier was trained in a similar fashion using the `train_data` and then subsequently evaluated using the `test_data`. Predictions were again observed using a confusion matrix as shown in Figure 7. The resulting `ROC-AUC` and `PR-AUC` metrics used for analysis are then displayed in Table 6.

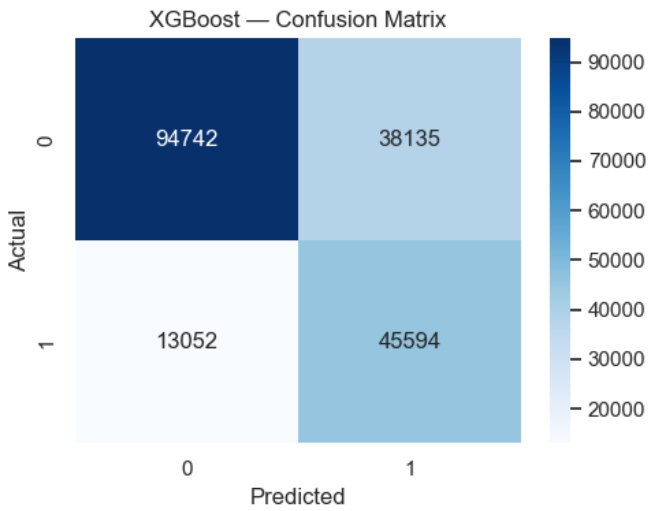


Figure 7. Confusion matrix for predictions from XGBClassifier.

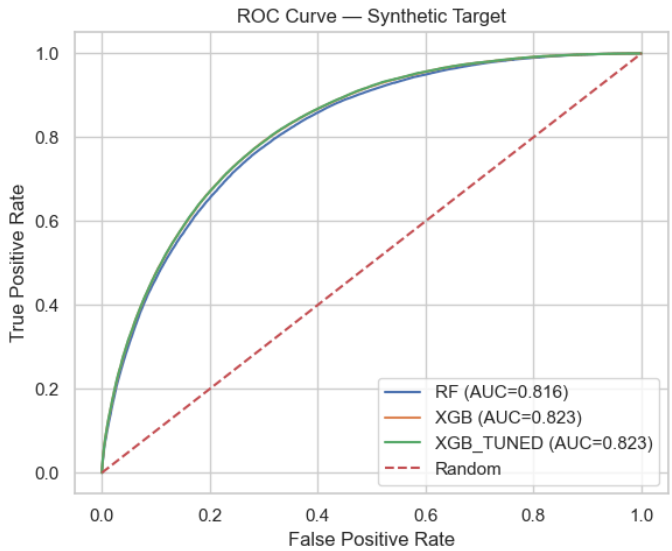


Figure 8. ROC curve for RFClassifier, XGBClassifier, and tuned-XGBClassifier models

Table 6. XGBClassifier Metrics Results

Metric	Synthetic Value	Original Value
ROC-AUC	0.502100	0.311300
PR-AUC	0.823014	0.659597

5.4. Hyperparameter Tuning

The decision to use the `XGBClassifier` model was made due to the slight increase in performance over `RFClassifier`. To perform hyperparameter tuning of the `XGBClassifier` model, a custom script was used to define a random parameter grid and evaluate the corresponding model performance across a validation set of the `train_data`. The parameters tuned were `n_estimators`, `learning_rate`, `max_depth`, `subsample`, `colsample_bytree`, and `reg_lambda`. After initializing the parameters and training the model accordingly, the set of best parameters (listed in Table 7) was determined and subsequently saved to `xgb_best.json` for use in final model evaluation.

Table 7. XGBClassifier Hyperparameter Tuning Best Parameters

Parameter	Value
n_estimators	100
learning_rate	6
max_depth	0.05
subsample	0.8
colsample_bytree	0.9
reg_lambda	1.0

5.5. Final Model Evaluation

After hyperparameter tuning was completed, final model evaluation was performed using the best hyperparameters. While introducing a weak-signal target in the data yielded significant improvement of model performance in both the RFClassifier and XGBClassifier models over standard baseline, hyperparameter tuning did not greatly increase these results. Minimal performance increase was seen utilizing the hyperparameters in the model. The ROC curve in Figure 8 shows the minimal increase in the tuned-XGBClassifer model over the original RFClassifier and XGBClassifer models.

6. Discussion

Initial modeling of the original target yielded poor, near-chance predictions and served as the basis for the generation of the synthetic target. As a result of the target transformation, the pipeline was able to generate predictions significantly better than random across all models due to the recovered weak-signal. In comparing the results of the two classification models on the synthetic target, the `XGBClassifier` slightly outperformed the `RFClassifier`, as evidenced by the edge in both `ROC-AUC` and `PR-AUC` metrics. In tuning the hyperparameters of the `XGBClassifier` model using the custom scripting on the validation set, the performance was enhanced minimally, improving by only $< 0.05\%$. The greatest advantage in our experiments proved to be the generation of the weak-signal synthetic target.

7. Conclusion and Future Work

Overall, the project yielded a reproducible, automated workflow that is capable of predictive modeling. This was achieved through initialization of a repository tracked with DVC and the replicable integration of the data via GCS. Furthermore, a predictive target was identified following automated data exploration, and a staged pipeline was created to perform predictive analysis of the target. Hyperparameter tuning was subsequently performed on the model pipeline, and model training and re-evaluation were performed using appropriate metrics. Completion of these tasks culminated in project success within the defined scope. Exploring the relevance of available predictors in prosthetic success to generate clinically-interpretable results could serve as a basis for a future study utilizing real-world data, with the workflow built in this project as a proposed tool for the analysis.

8. Reflection

In reflection, several lessons were learned throughout the project. The first learning revolved around knowing early the potential or limitations of your dataset. Similarly, ensuring the data is properly explored and analyzed prior to attempting any model planning. A bonus insight that came from these realizations was the opportunities that target engineering may offer in no-signal or weak-signaled data. Learning about several strategies to counteract this limitation, especially in synthetic data, provides options for different handling in future applications–especially if identified early in a project.

Practicing with DVC and similar replication tools (e.g. `uv`) enabled a better understanding of leveraging automated workflows. Despite the effort and preparation that went into staging the workflow, the

efficiency created on the backend with the reproducibility proves worthwhile. Given the real-world applicability of utilizing workflows, this project provided a good opportunity to practice these skills in an end-to-end flow.

References

[1] J. M. Czerniecki, A. P. Turner, W. A. Latzka, and B. J. Darter, "Mobility changes in individuals with dysvascular lower-extremity amputation secondary to peripheral artery disease and/or diabetes: A prospective cohort study", *Archives of Physical Medicine and Rehabilitation*, vol. 93, no. 9, pp. 1508–1515, 2012, Study describing ambulation changes in individuals with dysvascular lower limb amputation. DOI: 10.1016/j.apmr.2012.02.018.

[2] E. M. Balk et al., "Lower limb prostheses: Measurement instruments, comparison of component effects by subgroups, and long-term outcomes", Agency for Healthcare Research and Quality (US), Brown Evidence-based Practice Center, Rockville, MD, Comparative Effectiveness Review No. 213 18-EHC017-EF, 2018, Report reviewing evidence on measurement instruments and component effects in lower limb prostheses, highlighting evidence gaps in prosthetic component selection. [Online]. Available: https://www.ncbi.nlm.nih.gov/books/NBK531523/.

[3] D. C. Morgenroth, E. G. Halsne, A. W. Henderson, and D. C. Norvell, "Future directions in prosthetic component research and clinical prescription: A precision rehabilitation and patient-centered care approach", *Archives of Physical Medicine and Rehabilitation*, vol. 105, no. 12, pp. 2396–2399, 2024, Editorial discussing limitations of current evidence surrounding prosthetic component prescription and selection, proposing research directions for optimization. DOI: 10.1016/j.apmr.2024.08.016.

[4] N. Vanicek, C. Huang, F. Davie-Smith, and C. J. Hofstad, "Prescription of prosthetic ankle-foot mechanisms after major lower limb amputation: Cochrane protocol", *Cochrane Database of Systematic Reviews*, vol. 2025, no. 8, 2025, Protocol outlining systematic assessment of ankle-foot mechanisms for lower limb prosthesis users with emphasis on clinical prescription. DOI: 10.1002/14651858.CD016076.

[5] E. J. Earley, M. E. Awad, D. W. Milius, D. H. Melton, B. M. Gaffney, and J. W. Stoneback, "Consensus, recommendations, and guidelines for prosthetic limb prescription: A scoping review", *Journal of Prosthetics and Orthotics*, Publish Ahead of Print. December 25, 2025, Review summarizing available literature on prosthetic component prescription, including the role of clinician decision-making in component selection. DOI: 10.1097/JPO.0000000000000613.

A. Repository

All files and information related to the project are housed within the project respository, housed within the GitHub platform. The specific location for the project repository is:

```
https://github.com/wlgagli26/dsci_611_prosthetic_outcomes
```

The project requires the use of the `uv` tool in Python and is tracked utilizing `DVC`. To reproduce the contents of this report, use the `README.md` file to find instructions for running the workflow. The stages for the pipeline can be found in Appendix B along with the corresponding stage descriptions. The directory structure within the repository is found in Table 8.

Table 8. Repository Directory Structure

Title	Location	Comments
DVC Tracking	`/.dvc`	Local DVC files and configurations
Visual Studio Code	`/.vscode`	Project workspace settings
Data	`/data`	Raw and processed data
Models	`/models`	Model files
Model Metrics	`/metrics`	Metric results of experiments
Report Notebooks	`/notebooks`	Generated notebooks for project deliverables
Tuning Parameters	`/params`	Files with tuning parameters
Predictions	`/preds`	Files with modeling predictions
Source Files	`/src`	Project scripts
README.md	`root`	Project information and instructions
DATA.md	`root`	Data information and description
dvc.lock	`root`	DVC version locking (auto-generated for reproducibility)
dvc.yaml	`root`	DVC pipeline definitions
uv.lock	`root`	uv package locking (auto-generated for reproducibility)
pyproject.toml	`root`	Python configuration
.pre-commit-config.yaml	`root`	Pre-commit configuration checks
.gitignore	`root`	Git version control (ignore files)
.gitattributes	`root`	Git version control (configuration)
.dvcignore	`root`	DVC version control (ignore files)
.editorconfig	`root`	Code formatting
.dprint.json	`root`	Code formatting

B. Pipeline Stages

Table 9. Pipeline Stages

Stage	Description
`prepare_data`	Acquire and clean data
`split_data`	Split data into train/test sets
`train_rf`	Train RandomForest model
`train_xgb`	Train XGBoost Model
`tune_xgb`	Perform hyperparameter tuning of XGB
`train_xgb_tuned`	Re-train XGB with tuned parameters
`evaluate_models`	Perform model evaluation
`evaluate_report`	Generate evaluation report

C. AI Disclosure

Outside of grammatical checks and assisting the generation of BibTex citations, AI was not utilized to complete any deliverable for the project.